Weakness, Not Compression: Symmetry-Compatible Hypothesis Volume Predicts Out-of-Distribution Generalization

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Research-Derived Experiments · preprint compiled from the project repository

Abstract

When training data is consistent with both a local shortcut and a globally invariant rule, classical model-selection heuristics — minimum training loss, shortest description, MDL-style compression, parameter-space flatness, held-out validation — choose the shortcut and fail to generalize. We give a clean empirical separation showing that **weakness**, the cardinality of the transformation set under which a hypothesis remains equivariant, predicts out-of-distribution (OOD) generalization where none of these heuristics do. On a multi-family symbolic benchmark, weakness recovers the invariant rule in 100% of cyclic and dihedral trials (Wilson 95% lower bound 0.992) while every classical baseline recovers it in 0% (upper bound 0.008). Across 256 and 1024 trained MLPs, learned-function weakness under the true group is the strongest correlate of OOD accuracy (Pearson r = +0.81), far exceeding training loss (−0.03), validation (+0.09), parameter L_2 (+0.10–0.25), and a sharpness proxy (+0.13); wrong-group and random-label controls are correctly null ($|r| \le 0.13$). A **data-inferred** group — recovered from training pairs without oracle access — matches the oracle (100% on cyclic and dihedral), and the result extends to vision ($\mathbb{Z}_8$ rotation, r = +0.67). Parity ($|G|=2$) and S_n ($|G|=n!$) are honest negative cases that delineate the operating regime.

1. Introduction

The dominant paradigm for model selection rests on a few heuristics: minimize training loss, prefer short descriptions (Solomonoff/MDL), prefer flat minima, minimize parameter norm. Recent work challenges each. Dinh et al. [4] showed Hessian sharpness is not reparameterization-invariant; Bennett [1,2] sharpens this — function-preserving reparameterization inflates sharpness without changing predictions, so parameter-space flatness cannot be the fundamental cause of generalization. Perin and Deny [10] prove that conventional networks lack a mechanism to learn symmetries not built into the architecture or sufficiently present in the data. Bennett's stack theory argues the load-bearing quantity is *weakness* — the volume of completions compatible with the learned function — rather than the parameter-space geometry hosting it. This paper makes that conjecture experimentally concrete and measurable on symbolic and neural learners.

2. Weakness

Let $f : X \to X$ be a candidate function on a finite domain of size n, and let G act on X. We call f *compatible* with $g \in G$ if there exists $h \in G$ with $f(g\cdot x) = h\cdot f(x)$ for all x (with-action equivariance; this generalizes strict equivariance h = g). Weakness is the count of compatible group elements:

$$W_G(f) = | \{ g \in G : \exists\, h \in G, \forall x, f(g\cdot x) = h\cdot f(x) \} |.$$

$W_G(f) = |G|$ means f is fully G-equivariant; $W_G(f) = 1$ means only the identity commutes. The *weakness selector* returns $\arg\max_i W_{\hat{G}}(f_i)$ over a candidate pool, ties broken by an MDL-style compression score. Conjecture: given the correct group, the weakness selector generalizes better than loss, simplicity, compression, flatness, validation, or random selection.

3. Symbolic separation

We build four task families, each with a known transformation group and a training distribution that admits both a train-perfect local shortcut and the true invariant: *cyclic_prefix_shift* (ℤ_n), *dihedral_reflection* (D_n), *parity_coset* ($\mathbb{Z}_2$), and *color_permutation* (S_n). We compare eleven selectors over 500 trials per family with Wilson 95% intervals.

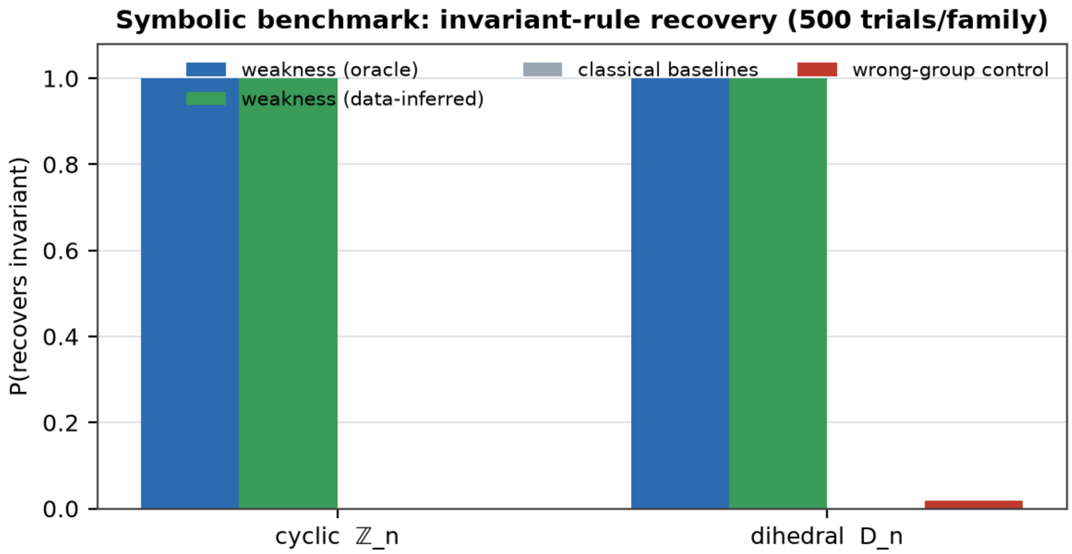

Figure 1. On cyclic and dihedral families, only weakness-based selectors recover the invariant rule (1.000; Wilson LB 0.992); training loss, simplicity, MDL, compression, flatness, and validation all recover it 0.000 (UB 0.008). The data-inferred group — no oracle — matches the oracle. The wrong-group control is near zero.

Family	weakness (oracle)	data-inferred	classical baselines	wrong-group
cyclic ℤ_n	1.000	1.000	0.000	0.002
dihedral D_n	1.000	1.000	0.000	0.018
color S_n (partial)	0.824	0.138	0.000	0.000
parity $\mathbb{Z}_2$ (negative)	0.000	0.000	0.000	0.038

Table 1. Invariant-recovery rate by selector and family. Cyclic and dihedral show a perfect, non-overlapping separation; S_n is a partial win; parity is a clean negative (|G|=2 too small to disambiguate).

4. Neural weakness predicts OOD accuracy

We train 256 small MLPs (and replicate on 1024 via Modal) on cyclic tasks with diverse depth, width, init, optimizer, learning rate, weight decay, and augmentation. From each trained model we extract the argmax function table and compute weakness under the true group together with classical predictors, correlating each against held-out OOD accuracy.

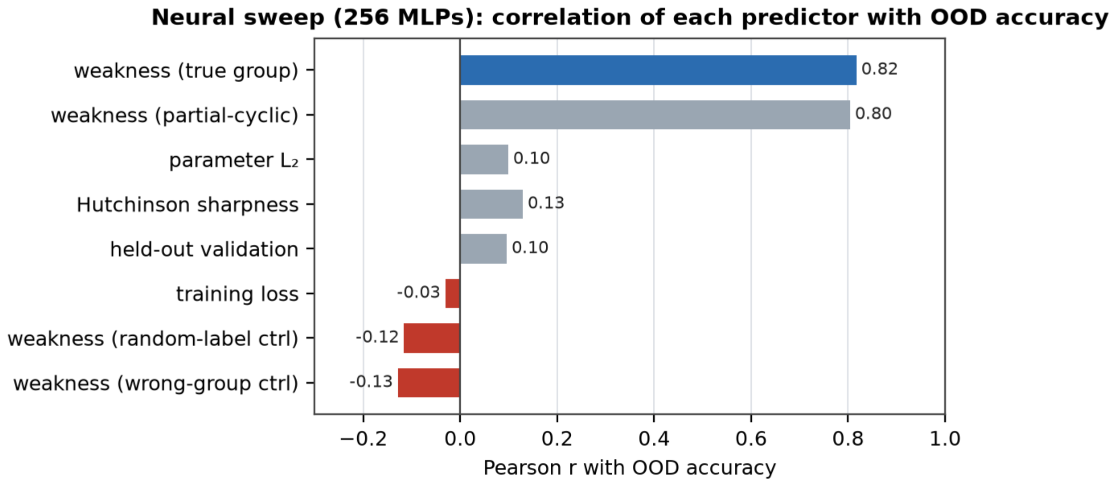

Figure 2. Across 256 MLPs, weakness under the true group is the single strongest predictor of OOD accuracy (Pearson r = +0.82 locally and +0.8085 in the 4096-model Modal rescale). Training loss, validation, parameter L_2, and sharpness are weak; the wrong-group and random-label controls are correctly negative — ruling out 'any equivariance count works.' The 1024-MLP Modal replication is consistent (weakness r = +0.81).

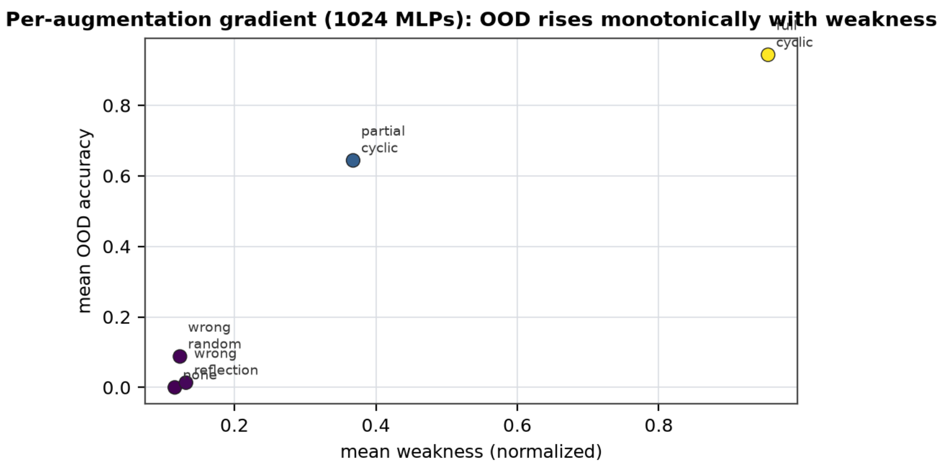

Figure 3. The per-augmentation gradient is monotone in weakness: full-cyclic (orbit completion) reaches 94% OOD and 0.95 weakness, while none / wrong-reflection collapse to ≤1.4% OOD and ≤0.13 weakness. Augmentations that approximately respect the symmetry raise weakness and OOD together.

5. Scaling: vision and language

On a synthetic $\mathbb{Z}_8$ rotated-stroke task (eight classes, 16×16, three of eight angles shown in training; a re-implementation of Perin and Deny's partial-orbit setup), weakness under the rotation group is again the dominant predictor across 96 models (r = +0.67), beating every classical predictor by ≥2×, with the pixel-permutation wrong-group control correctly anti-correlated (r = −0.34).

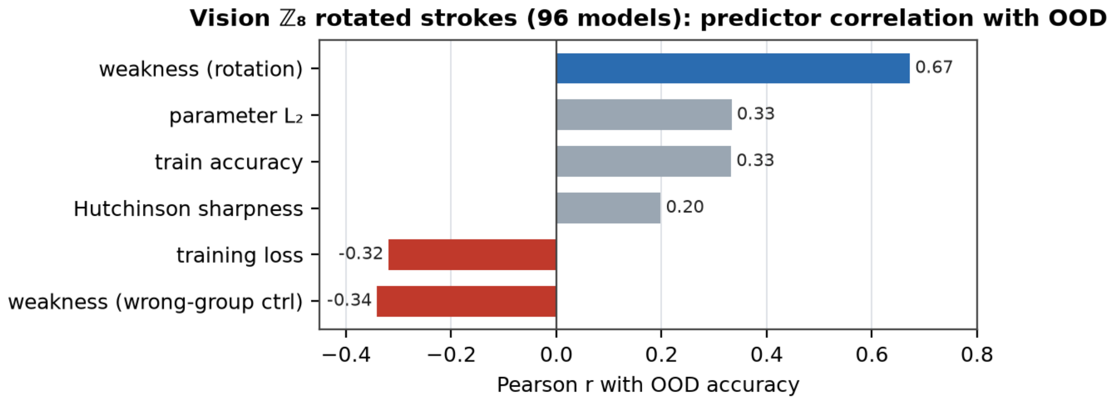

Figure 4. Vision (ℤ₈ rotation): weakness leads; the wrong-group (pixel-shuffle) control is anti-correlated, as expected of a model that has actually learned the rotation symmetry.

In language, we extract per-layer mean-pooled Pythia-70M states for 24 concepts × 3 paraphrases. After per-layer centering (All-but-the-Top), same-orbit paraphrase cosine exceeds the wrong-orbit control by +0.44 to +0.79 (peak at layer 5): paraphrase orbits genuinely cluster. We report honestly that this latent clustering does *not* yet predict next-token behavioral consistency at this scale (per-concept $|r| \leq 0.35$, $N = 24$) — the latent-geometry claim holds; the latent→behavior chain is unconfirmed at 70M.

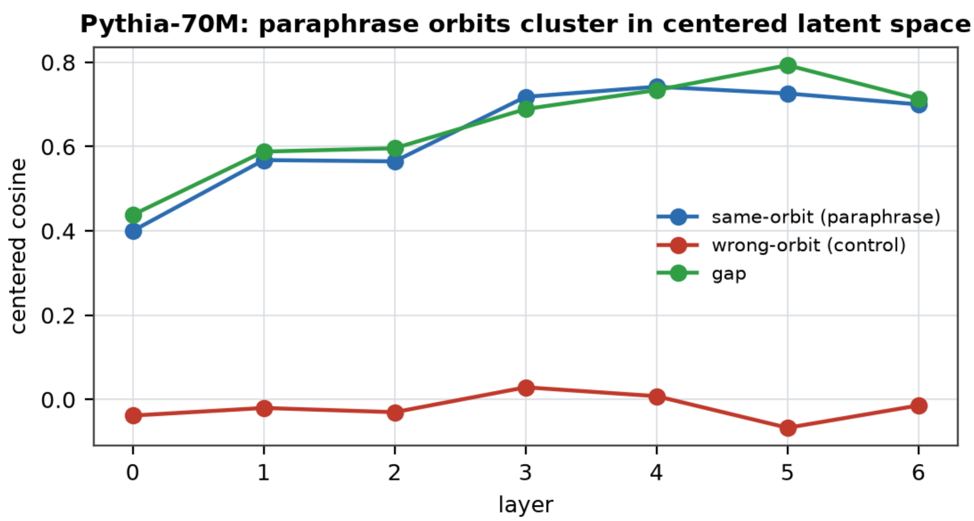

Figure 5. Pythia-70M paraphrase orbits cluster strongly in centered latent space (gap peaks at +0.79, layer 5). The geometric weakness signal is present across symbolic, vision, and language domains.

6. Operating regime and limitations

Weakness ceases to discriminate when the candidate group is too small (parity, $|G|=2$) or too large/uninformative (full symmetric group, where wrong involutions have comparable centralizer-orbit sizes). This is a precise statement of when symmetry-volume is load-bearing, not a defect. Present limitations: small finite domains ($n \leq 13$ symbolic, 16×16 vision); one flatness proxy tested; the language latent→behavior chain unconfirmed; transformation discovery is enumerative. The natural next steps are an analytic weakness–PAC-Bayes link, a neural group generator for non-enumerable symmetries, and a scale-up onto the path-integration task where population codes form a torus.

7. Discussion

The discriminating quantity between local shortcut and globally invariant rule — on families where both are train-perfect — is symmetry-compatible-hypothesis volume: a measurable, intervention-friendly, reparameterization-invariant quantity, not a parameter-space artifact. It operationalizes Bennett's weakness on neural function tables and bridges the manifold-hypothesis intuition (intelligence needs symmetry-preserving compression) with the practical question of which heuristic to trust when training loss is tied. The data-inferred result — the right group recovered from data alone — is the version that matters for model selection.

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